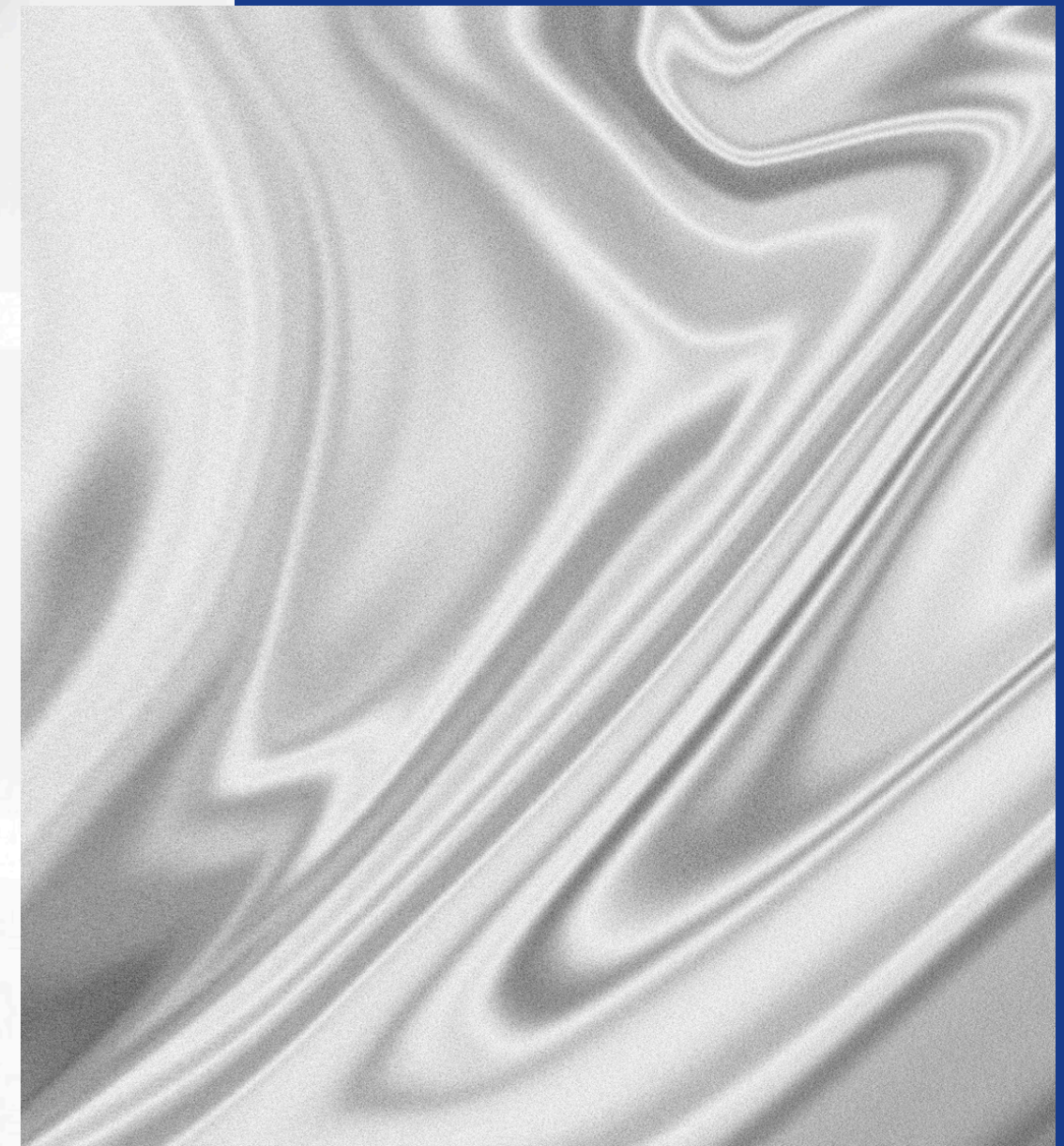


Group 2 CPE 3104 MICROPROCESSORS

Automatic Marker Refiller

Group Members:

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Problem

Importance of Ink:

The use of ink is fundamental to producing images, text, and designs, whether through traditional tools like pens and brushes or in modern printing processes such as letterpress and lithography.

Ink is a complex medium, often containing solvents, pigments, dyes, resins, and various other additives that govern its flow, thickness, and final appearance.

Identified Problem:

Manual marker refilling exposes users to harmful solvents and pigments, creating potential health risks. It also leads to collateral issues such as spills, stains, and workspace contamination.

An Automated Marker Machine (AMM) eliminates these hazards by removing direct contact with ink materials and minimizing human error during the refilling process.



Features

Use Case	Description	Trigger
Refill Marker	Determines required ink volume via sensor reading, activates pump for a calculated duration, and monitors for completion.	Marker inserted and detected by Limit Switch.
Display Status	Provides real-time feedback on system state, remaining ink, and error codes.	Continuous (Programmed I/O loop).
Emergency Stop/Halt	Immediately halts pump and motor movement to prevent spills or mechanical damage	Limit Switch interrupt, or software-detected error (e.g., out-of-range sensor reading).
Self-Test	Checks sensor function and motor driver connection during startup.	System initialization.

Hardware Design

Hardware Components used in Proteus:

- Power Supply
- 8086 microprocessor
- LCD Display (LM044L)
- Marker Slot (4 SW)
- Motors (4)
- Button (2)
- 74ls245 IC (2)
- 74ls373 IC (2)
- 74ls138 IC w/ NOT gate
- 8255A (2)
- 8259
- Potentiometer
- ADC0808



Software Design

- The design follows a polling-based main loop structure, which continuously checks the system's state and sensor inputs. This main loop handles the core logic: reading the ink-level sensor (ADC), checking the start button, detecting which marker slots are occupied, and activating the corresponding refill pumps for a fixed duration.
- The START routine sets up the data and stack segments, programs the two 8255 PPIs for I/O expansion to control the LCD, ADC, and pumps. And configures the 8259 PIC. It establishes an Interrupt Vector Table (IVT) entry at 200H to map an external hardware interrupt to the ISR1 routine.

Software Design

- Ink Level Sensing (ADC): The routine controls the ADC0808. It reads the ink level and compares it against a threshold. If the level is LOW, the START_FLAG is cleared, preventing refilling.
- Marker Detection & Refilling: If the START_FLAG is SET, the program polls Port 2C, Bits 0-3 to detect which marker slots are occupied. It uses marker-specific flags to register the presence of a marker. The MOTOR_CONTROL section then combines these flags into a bitmask sent to Port 2A to activate the corresponding pumps. The pumps run for a fixed, software-controlled 5-second delay.

Software Design



- Message Management: Utility routines like `LCD_CLEAR`, `LCD_SEND_COMM`, `LCD_SEND_CHAR`, and `LCD_DISP_TEXT` abstract the low-level hardware communication, allowing the main logic to easily display status messages at various points in the operation.